

Cycles 43-45 Report — Post-Merge Integration of Cycle-37 Fanout
(Fork 675abd086911)

2026-08-29

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Cycles 43-45 Report — Post-Merge Integration of Cycle-37 Fanout (Fork 675abd086911)

Abstract

Cycles 43-45 constitute the root-conductor’s post-merge integration of the three-branch cycle-37 fanout on fork 675abd086911. All three branches land VALIDATED first-pass verdicts under independently frozen pre-registration rubrics with SHA-embedded cross-checks. The merge closes the operator-priority M-RECREATE-1/first-real-audio at **RECREATION_LANDS** (first end-to-end recreation on a real rated song), discharges the c36 SB3-statistic-degeneracy blocker at _manager/ear-sb3-statistic-degeneracy-fallback-statistic **F1_ADOPTED** (unblocking M-EAR-1/real-label-training-v1 for c38), and closes M-GEN-1/palette-driven-batch-v4 at **PARAM_MOVES_AUDIO** with 28/28 cross-salt SHA distinctness.

Merged Verdicts (Three Branches)

Branch	Milestone	Verdict	Rubric SHA-256 (leading 16)
Clone 0	M-RECREATE-1/first-real-audio	RECREATION_LANDS	78c61c5dbf61492f...

Branch	Milestone	Verdict	Rubric SHA-256 (leading 16)
Clone 1	<code>_manager/ear-sb3-statistic-degeneracy-fallback-statistic</code>	F1_ADOPTED	0ba2be8b18ba5f09...
Clone 2	<code>M-GEN-1/palette-driven-batch-v4</code>	PARAM_MOVES_AUDIO	bd361e3e50af9dc3...

Each rubric SHA embedded verbatim in its verdict JSON and asserted by dedicated test. Byte-determinism $\times 2$ verified on every deterministic artefact family across all three branches.

Clone-0 — M-RECREATE-1/first-real-audio (First End-to-End Real-Song Recreation)

- **Verdict:** RECREATION_LANDS. Effects layer narrows `mel_l1_db` by **+5.906 dB** (31.229 $\rightarrow$ 25.323); clears the LANDS gate.
- **Song selection:** SHA-256 tiebreak $\rightarrow$ `corpus/ratings/7/016_LOCAL_05_02.mp3` (band 7, 1,659,745 bytes, SHA 069ebba2...5048); trimmed to first 30 s per rubric duration bound.
- **8-stage pipeline** (80.55 s total wall, run 1): `decode` $\rightarrow$ `chunker` $\rightarrow$ `tagger sidecar` $\rightarrow$ `htdemucs` (4 sources, `shifts=0`, `overlap=0.25`) $\rightarrow$ `basic-pitch` $\times 3$ (quarantined venv) $\rightarrow$ `score merge` $\rightarrow$ `fluidsynth bare-MIDI` (SF2 SHA 74594e8f...1cb0) $\rightarrow$ `cycle-9 DawDreamer effects` (read-only).
- **Byte-determinism $\times 2$** on all four deterministic anchors: `merged.musicxml`, `merged.midi`, `bare_midi.wav`, `effects.wav` (SHAs 95de5356...1592, 5cccca6c...c599, 0658c70f...39f4, 8974db22...ce7a).
- **Panels honest:** effects narrows `mel/env/loudness` (`mel_l1_db`, `rms_env_rmse`, `lufs_m_rmse_lu`); regresses on `spectral_centroid_rmse_hz` (+250.36) and VGGish (0.29457 $\rightarrow$ 0.33251) — Surge XT reverb+chorus adds spectral spread the mastered original doesn't have. Same content-dependent family disagreement cycle-13 characterised; the panel surfaces it rather than aggregating.
- **12 read-only upstream anchors byte-identical:** `chunker`, `tagger`, `sidecar_nonfactor`, `htdemucs runner`, `basic-pitch driver` + `_bp_call`, `score bridge`, `render_bare_midi`, `render_effects_layered` (cycle-9 chain locked), `texture panel`, `ear features`, `ear model`.
- **Preview-untrained-ear caveat prominent** per `data/recreate_v0/ear_score_untrained.json` sentinel; cites `c36 M-EAR-1/real-label-training-v0` $\rightarrow$ `EAR_v0_INSUFFICIENT`; MUST NOT influence recreation verdict.
- **Deviations disclosed:** (1) 30 s trim (pre-authorised by rubric); (2) stage-06 `pretty_midi fallback` (`mscore3 xml_to_midi` duration-quantization failure on real-audio-derived MusicXML; deterministic $\times 2$; c38 handoff #1 owns upstream fix).
- **Tests:** 18/18 PASS (structural 8; artefact 10).

Clone-1 — `_manager/ear-sb3-statistic-degeneracy-fallback-statistic` (F1_ADOPTED)

- **Verdict:** F1_ADOPTED. Aggregate 2.500 vs F3 2.480 ($\Delta = 0.020$ near-tie backup); F2 disqualified on T2 (FPR = 0.17 $>$ 0.10 on singleton_43).
- **Threshold results:** all three candidates pass T1 (detection ≥ 0.90 @ $\alpha=1.0$ repeat_55) = 1.00; F1 + F3 pass T2 (FPR ≤ 0.10 @ $\alpha=0$ singleton_43); all three pass T3 (0/100 SHA-256 mismatches $\times 100$ salts $\times 2$ fixtures).
- **F1 singleton-degeneracy invariant** (feature not bug, pre-registered rubric §10): F1 saturates at 2/3 on singleton corpora analytically (`SS_between == V_pool` when every

group has size 1). Downstream contract: F1-based leak test returns SB3_UNRESOLVED_SINGLETON_CORPUS on 43-song rated corpus — exactly the honest signal c36 clone-0's EAR_v0_INSUFFICIENT was asking for.

- **Anchor preservation** (5/5 byte-identical before $\Rightarrow$ after): data/ear/leak_test_summary.json, scripts/ear/{leak_test,synthetic_labels,stability_audit}.py, docs/ear_path_b_commitment.md.
- **Isolation**: no sidecar_nonfactor import, no i4_stratified import, no PRNG (AST + grep verified); interpreter guard on 7/7 new scripts.
- **Tests**: 20/20 PASS (exceeds ≥ 14 minimum). c36 blocker_manager/ear-sb3-statistic-degeneracy-on-singleton-artists-clone-0 transitions to superseded (by F1_ADOPTED) at merge.

Clone-2 — M-GEN-1/palette-driven-batch-v4 (PARAM_MOVES_AUDIO Deeper Perturbation)

- **Verdict**: PARAM_MOVES_AUDIO. **28/28 cross-salt bare_combined.wav SHA pairs distinct** across salts 0..7 (up from c36 clone-1's 3/3 across salts 0..2). Per-salt byte-determinism $\times 2$ all-PASS; panels 8-key finite on all 16 (8 salts $\times 2$ panels) comparisons.
- **v4 IQR > v3 IQR on all 8 numeric-family texture-panel keys** — deeper perturbation genuinely diversifies the audio, not just metadata.
- **Ships c36-deferred sfizz opcode-file-rewrite fallback**: extend_sfizz_opcode_rewrite.rewrite_sfz_to_temp rewrites target-opcode values in-memory before invoking sfizz_render, then restores file. Expands parameter_dict from 4×4 to 8×8 (fluidsynth gain/chorus/reverb/LP-cutoff/HP-cutoff; sfizz master_volume/cutoff/resonance). Salt range 3 $\rightarrow$ 8.
- **Additive-only surgical edit** to scripts/palette_render/render_stem.py: lazy import of rewrite_sfz_to_temp when parameter_dict contains cutoff or resonance. Signature unchanged. c33 path (parameter_dict=None) byte-identical.
- **Backwards-compat**: 3/3 c33 anchor SHAs byte-match (bass 6b9a5219..., other a2e5d058..., combined a8c1557c...) under parameter_dict=None.
- **In-cycle bug caught and fixed**: first batch run produced silent output (LUFS = $-\infty$) because opcode-rewritten SFZ was written to /tmp/*.sfz while sample=test_saw.wav remained relative; sfizz resolves samples relative to SFZ file, so /tmp/test_saw.wav did not exist. Fix: _absolutize_sample_paths rewrites sample=<relative> to absolute-from-source-directory (test 13). Second run: salt-0 combined LUFS = -16.85 (finite).
- **Anti-patterns locked**: c31 STILL_GAP + c35 A RENDER_FAILS unchanged; VST3 branches raise NotImplementedError on non-None parameter_dict. c9 chain / c13 batch-v2 / c15 i4_stratified / sidecar_nonfactor NOT imported; interpreter guard on every script.
- **Tests**: 21/21 PASS on test_palette_driven_batch_v4.py; c36 clone-1 regression test_palette_driven_batch_v3.py 20/20 PASS.

Merge-Time Ledger Concatenation

Per-clone shadow ledgers at /home/user/music-gen-instance/fork-675abd086911/clone-{0,1,2}/promise_ledger.jsonl concatenated via long_exposure.workspace_bootstrap.concat_clone_ledgers. Row counts:

Clone	Shadow rows (6 named + 2 housekeeping)	Convention
0	8	-clone-0 on infra families; substantive M-RECREATE-1/first-real-audio unsuffixed (5 rows)
1	8	-clone-1 on infra families; six named events under _manager/ear-sb3-statistic-degeneracy-fallback-statistic-clone-1/
2	8	-clone-2 on infra families; substantive M-GEN-1/palette-driven-batch-v4 auto-suffixed per c33+c36 v2 guard when merged from clone context

Canonical-hash concat (c27) deduplicates housekeeping events. Straightforward merge; no expected LedgerConcatError. No shared substantive M-* labels across clones; no filesystem conflicts (all new files under docs/recreate_v0_*, scripts/recreate_v0_*, data/recreate_v0_*, docs/ear_sb3_fallback_statistic_*, scripts/ear_sb3_fallback_*, data/ear_sb3_fallback_*, docs/palette_driven_batch_v4_*, scripts/palette_render_v4_*, data/palette_render_v4_*, tests_*, tools/stale/).

State-Machine Discipline (c29 Lemma Respected)

All three substantive milestones are peer sub-milestones under existing terminal-validated parents:

- M-RECREATE-1/first-real-audio — peer under M-RECREATE-1 (new milestone family; first end-to-end recreation on real audio).
- _manager/ear-sb3-statistic-degeneracy-fallback-statistic — peer under _manager/* (analytical rubric-design fix).
- M-GEN-1/palette-driven-batch-v4 — peer under M-GEN-1; NOT a child of terminal-validated palette-driven-batch-v{1, 2-sampler-diversified, 3} or batch-v{1..6}.

Zero validated → in_progress transitions attempted.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908; no refit.
- SHA-256 tiebreak; no PRNG; no sidcar_nonfactor / i4_stratified imports in analytical scripts.
- Interpreter guard /usr/bin/python3 on every new script across all three branches.
- Read-only anchors preserved: c6 feature cache + leak-test surface; c9 DawDreamer effects chain; c13 batch pipeline; c22 stability harness; c26 Path B commitment; c31 palette-v1 + palette_probe; c33 palette_render base + dawdreamer_state; c34 palette_v2; c35 palette_v2_render; c36 palette_render_v3.
- Rated audio egress at *.googlevideo.com: still 403; non-blocking probes emit at cycle top per usual. Recreation runs on the 43-song operator-delivered corpus (on-disk); no egress required.

- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses
UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).

Anti-Patterns Locked (5-Count Stable; c35 A #6 + c31 STILL_GAP Re-inforced)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted across any branch. c31 STILL_GAP / c35 A anti-pattern surface intact (VST3 branches remain NotImplementedError-quarantined in clone-2). c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged.

Cycle-38 Handoff (Priority Order)

Highest priority (operator-carried recreation line): 1. **M-SCORE-1/bridge-api on real audio** — fix the pretty_midi fallback: either add rational-duration snapping in merge_stems_to_score, or replace mscore3 in xml_to_midi with a music21 → mido pathway that tolerates raw durations, or accept the pretty_midi concat as second-class. 2. **M-RECREATE-1/first-real-audio/cross-song-batch-v0** — extend to $N \in \{3, 5\}$ songs (next-N by SHA-256 tiebreak: Tom Misch — Red Moon; Justin Bieber — YUKON Grammys; Dayme Arocena — La Rumba) on the same rubric to verify LANDS generalises across bands 4-7. 3. **Panel-disagreement summary field** — add panel_disagreement descriptor distinguishing mel-only-LANDS from full-panel-LANDS.

M-EAR-1 real-label-training-v1 (now unblocked by clone-1 F1_ADOPTED): 4. **Retire c6 $S = \max(S_{\text{model}}, S_{\text{resid}})$ line** in scripts/ear/leak_test.py in favour of F1 (anchor was READ-ONLY under c37 clone-1 scope; c38 owns the actual edit). 5. **Consume the SB3_UNRESOLVED_SINGLETON_CORPUS contract** (rubric §10, pre-registered): the F1-based leak test on the 43-song singleton-artist corpus MUST return SB3_UNRESOLVED_SINGLETON_CORPUS rather than a numerical detection percentile. 6. **Corpus scale is the leading candidate variable** (per c36 clone-0 close): within-artist corpus expansion, not chassis redesign (locked out per c22/c23/c25 anti-patterns).

Palette-render arc (post-clone-2 PARAM_MOVES_AUDIO deeper-perturbation): 7. **VST3 activation gated by c36 Branch C MIXED verdict** — candidates: M-DAW-SPIKE-1/dexed-only-vst3-tolerance-activation; M-DAW-SPIKE-1/vst3-envelope-tolerance-activation; or leave STILL_GAP. 8. **Thread lp_cutoff and hp_cutoff to fluidsynth CLI** — c38 can promote via synth.reverb.damp / synth.chorus.speed, or admit the fallback permanently. 9. **M-GEN-1/palette-driven-batch-v5** candidates: wider table (16×16); non-uniform value ladder; per-note MIDI CC automation.

Infra: 10. **Guard-refinement candidate:** c33 harness-clone-namespace-guard endswith check should match -clone-<digit>+/[^/]+ (any tail past clone-suffixed parent) as also “already namespaced” — avoids the cosmetic double-suffix; workaround at tools/stale/_fix_shadow_clone_ids.py. 11. **_infra/merge-report-path-fallback-convention** — fourth-observation of sandbox-write refusal pattern (c31/c34/c35/c36 all showed it); worth first-class codification.

Cumulative Progress

M-EAR-1 arc (unchanged operator posture; analytical branch closed by c37 clone-1):

Cycle	Milestone	Verdict
c22-c25	Path A chassis chain	insufficient (three-audit chain, anti-patterns locked)

Cycle	Milestone	Verdict
c26	<code>_manager/M-EAR-1-path-B-commit</code>	committed; three SBs frozen
c31	<code>armed-harness-fixture-reinforcement</code>	FIXTURE_READY
c36	<code>real-label-training-v0</code>	EAR_v0_INSUFFICIENT (first real-label fire)
c37 clone-1 (this merge)	<code>_manager/ear-sb3-statistic-degeneracy-fallback-statistic</code>	F1_ADOPTED
c38 (next)	<code>real-label-training-v1</code>	unblocked

M-RECREATE-1 arc opens: first end-to-end recreation on a real rated song landed RECREATION_LANDS. Cross-song batch expansion is the natural next step.

M-GEN-1 palette line — five-cycle mechanism-focused convergence chain:

Cycle	Milestone	Verdict	Structural Progress
c33	<code>M-TEX-1/palette-driven-bare-render</code>	PALETTE_MOVES_PANEL	single-song activation
c34	<code>M-GEN-1/palette-driven-batch-v1</code>	BATCH_SPREAD_COLLAPSED	dispatcher rule_id-invariant
c35	<code>M-GEN-1/palette-driven-batch-v2-sampler-diversified</code>	SPREAD_STILL_COLLAPSED	render_stem API surface never consumed
c36	<code>M-GEN-1/palette-driven-batch-v3</code>	PARAM_MOVES_AUDIO	pinned_state additive parameter_dict kwarg (3-salt)
c37 clone-2 (this merge)	<code>M-GEN-1/palette-driven-batch-v4</code>	PARAM_MOVES_AUDIO (8-salt, 28/28 distinct, IQR > v3 on all 8 keys)	

Pattern durability: **six consecutive cycles** of rubric-first pre-registration discipline (c26-c37). Every cycle since c26 has committed a verdict rubric before analysis, with rubric SHA embedded verbatim in verdict JSON and a git-mtime-order test asserting it. Zero after-the-fact rubric edits. Recommend codification into plan-of-record standing practice.

c29 state-machine lemma respected: every c37 fanout branch is a NEW peer sub-milestone; ledger topology stays a DAG.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard (c36 v2): infra families `-clone-<k>-suffixed`, substantive `M-*` unsuffixed (auto-suffixed at merge from clone context per c33+c36 v2 guard). Cosmetic double-suffix quirk on already-namespaced parents surfaced for the first time; workaround archived at `tools/stale/_fix_shadow_clone_ids.py`; lint-level fix owed by future infra cycle.

M-EAR-1 armed-harness Path B now armed-and-fired with honest negative finding + statistical fallback in place. **Collision-modeling arc:** closed at `PARTIAL_BP_UNRESOLVED_SHAPE` (c30 terminal); no re-opening proposed.

Merge state: cycle-37 fanout fully absorbed; all three branches ready for concat_clone_ledgers; integration tests green across the fork; 0-ERROR promise_check (advisory orphan-artifact WARNs will clear on housekeeping event absorption). Campaign is ready for cycle 38.

[END OUTPUT]